Semester :- Sem VI
Branch :- Computer science and engineering
Subject :- Design and analysis of algorithm

UNIT-I

List and explain in detail different general means for improving the efficiency of algorithms.

How inefficiency is caused due to late termination? Explain with suitable example.

List and explain the different aspects in constructing a loop.

Explain the concept of recursion with suitable example.

Explain recursive & non –recursive programming with reference to efficiency.

Explain

i) Execution Trace

ii) Regular Expressions.

Write regular expression for execution trace of following algorithm and convert it in to iterative.

```
If(f(n)) then
        s(n);
        rec(g(n));
end
```

Convert the following Fibonacci series from recursive implementation to iterative one.

$$Fib(n)=\begin{cases} n & \text{if } n=0 \text{ or } n=1 \\ Fib(n-2)+Fib(n-1) & \text{if } n>1 \end{cases}$$

List ard explain different algorithm design strategies.

An algorithm takes 0.5ms for input size 100. How long it will take for input size 500 if the running time is the following.

Linear ii) Quadratic iii) O(N log N) i) Cubic

Explain Big-Oh, Omega & Theta notations.

Explain with an example how to remove redundant computations outside loops.

Explain with an example conversion of an iterative function to a recursive function.

Write recursive factorial function and convert it into iterative one.

What is the importance of Asymptotic notations? Explain it with the help of big - oh and theta notation.

Why problem size is used as measure for time complexity? Explain.

How referencing of array elements affect efficiency of algorithms? Explain with suitable example.

Explain Big-oh notation in detail.

UNIT-II

Explain Strassen's Matrix Multiplication method with example.

List and explain the different types of test performed on triangulation.

Explain the different types of convex hull algorithms.

What is divide and conquer? Explain it’s advantages and disadvantages.

Explain Strassen matrix multiplication technique. What is it's time complexity?

What do you mean by triangulation? Explain different applications of triangulation.

Write an algorithm for merge sort. Explain how divide and conquer strategy is applicable to it. Also analyse the algorithm.

Explain advantages and disadvantages of divide and conquer. List the factors by which running time of divide and conquer algorithm is affected.

What is divide and conquer ? Explain with suitable example.

What the concept of closest pair. Explain the closest pair algorithm.

Expalin Binary search algorithm using recursive and iterative version.

What is divide and conquer strategy? Write control abstraction for it.

Explain the giftwarpping algorithm?

Explain with help of suitable example that convinces the power of divide-and-conquer strategy.

Explain multiplication algorithm based on divide-and-conquer strategy and analyse it’s performance.

What is Divide and conquer? Explain it with the example of finding power of given number.

Write the algorithm for Binary Search. How it is solve using divide and conquer strategy? explain its Best , Average and Worst case time complexity.

Explain Merge sort algorithm with suitable example.

Where does the divide and conquer strategy fail? Explain the class of problems for which it is unsuitable.

UNIT-III

Explain what greedy algorithms are and write its abstract algorithm.

Explain the terms feasible solution, optimal solution and objective function?

What is greedy algorithm? Explain with help of making change example. List its general characteristics .Also write its (greedy algorithm) function.

Explain knapsack algorithm to find an optimal solution to the following instance of knapsack problem if n=3 M=20 (P_1, P_2, P_3)=(25,24,15) and (W_1, W_2, W_3)=(18,15,10).

Find an optimal solution to the knapsack instance : n=7 ,M=15, $(P_1, P_2, P_3 \ldots P_7)$=(10,5,15,7,6,18,3) and $(W_1, W_2, W_3 \ldots W_7)$=(2,3,5,7,1,4,1).

Explain knapsack algorithm to find an optimal solution to the following instance of knapsack problem if n=5 M=100 $(P_1, P_2, P_3 \ldots P_5)$=(20,30,66,40,60) and $(W_1, W_2, W_3 \ldots W_5)$=(10,20,30,40,50).

Find solution to the scheduling problem when n=6, P_i=(20,15,10,7,5,3) and D_i=(3,1,1,3,1,3)

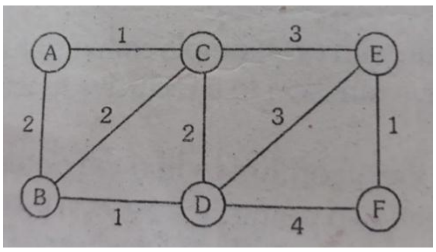
Find solution to the scheduling problem when n=7, P_i =(3,5,20,18,1,6,30) and D_i=(1,3,4,3,2,1,2)

Find solution to the scheduling problem when n=4, P_i=(100,10,15,27) and D_ii=(2,1,2,1)

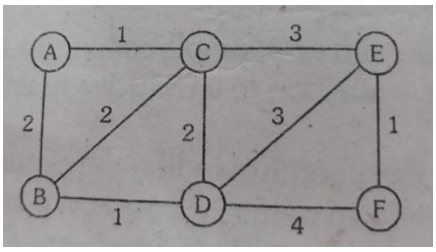
Find solution to the scheduling problem when n=4, P_i=(30,35,20,25) and D_i=(2,1,2,1)

What are minimum spanning tress, List the characteristics and approaches of greedy algorithm to obtain the minimum spanning trees.

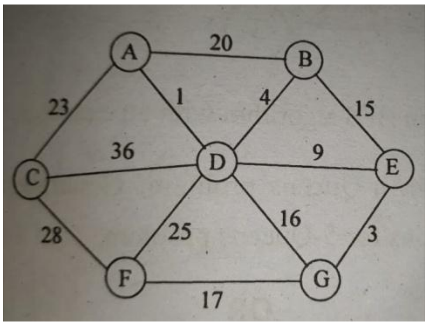
Simulate the Prim's algorithm for the following graphs



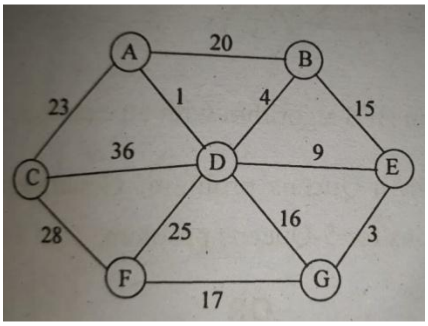
Simulate the Kruskal's algorithm for the following graphs



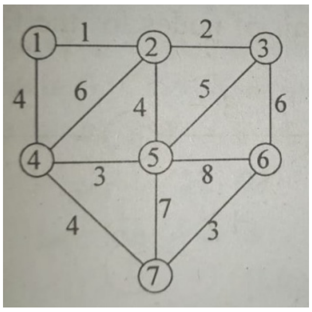
Simulate the Prim's algorithm for the following graphs



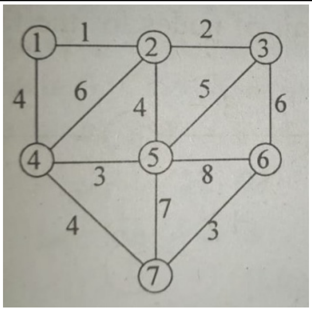
Simulate the Kruskal's algorithm for the following graphs



Simulate the Prim's algorithm for the following graphs



Simulate the Kruskal's algorithm for the following graphs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UNIT-IV

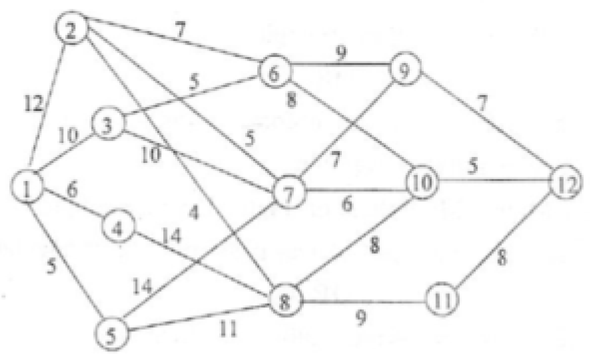
UNIT-4

1) Explain the term Principle of optimal with suitable examples.

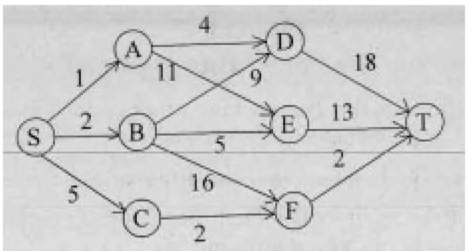
2) Distinguish between greedy methods and dynamic programming. Give its advantages and disadvantages.

3) Explain optimal polynomial triangulation with suitable examples.

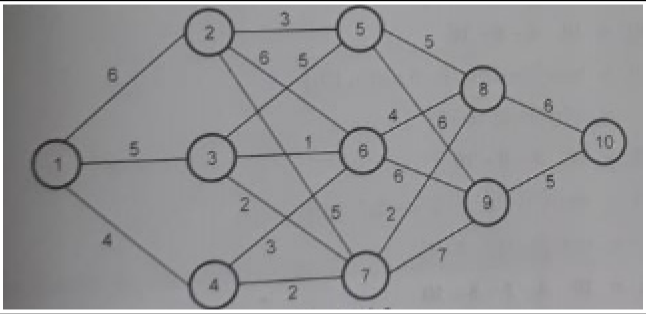
4) What is the multistage graph problem? Find the shortest path for following a multistage graph using dynamic programming.



5) Find the shortest path for following multistage graph using dynamic programming:



6) What is the multistage graph problem? Find minimum cost path from s to t in the following multistage graph.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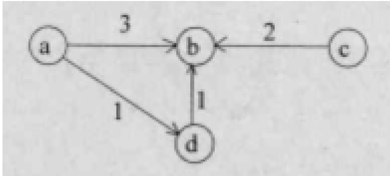
7) Explain Floyd-Warshall algorithm to find the shortest path with an example.



8) Explain Floyd's algorithm for computing all pairs shortest paths. Also find the matrix D that gives the length of shortest path between pairs of nodes for the following directed graph.

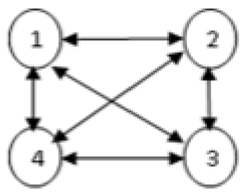
$$D0=L=\begin{pmatrix} 0 & 5 & \infty & \infty \\ 50 & 0 & 15 & 5 \\ 30 & \infty & 0 & 15 \\ 15 & \infty & 5 & 0 \end{pmatrix}$$

9) What is a traveling salesperson problem? Solve it for the directed graph given below, the edge lengths are indicated by the matrix.

$$\begin{bmatrix} 0 & 2 & 9 & 10 \\ 1 & 0 & 6 & 4 \\ 15 & 7 & 0 & 8 \\ 6 & 13 & 12 & 0 \end{bmatrix}$$

What is a traveling salesperson problem? Solve it for the directed graph given below, the edge lengths are indicated by the matrix.

$$\begin{pmatrix} 0 & 10 & 15 & 20 \\ 5 & 0 & 9 & 10 \\ 6 & 13 & 0 & 12 \\ 8 & 8 & 9 & 0 \end{pmatrix}$$



11) Explain how a chain matrix multiplication algorithm can be solved using a dynamic programming approach.

12) Explain longest common subsequence (LCS). Find the sequence for following string:

a) X_i= A, B, C, B, D, A, B and Y_j= B, D, C, A, B, A

b) X=<A,B,C,B,A> and Y=<B,D,C,A,B>

c) A=<X,Y,Z,Y,T,X,Y> and B=<Y,T,Z,X,Y,X>

13) Use dynamic programming strategy to calculate product of four matrices ABCD , where A is 5*4,B is 4*6,C is 6*2 D is 2*7.

14) Use dynamic programming strategy to calculate product of four matrices ABCD , where A is 13*5,B is 5*89,C is 89*3 D is 3*34.

15) Explain single source shortest path with example.

16)Explain the chained matrix multiplication algorithm for dynamic programming. State how the algorithm works to calculate the product of four matrices M1 is 10 * 20, M2 is 20*50, M3 is 50 *1 and M4 is 1*100.

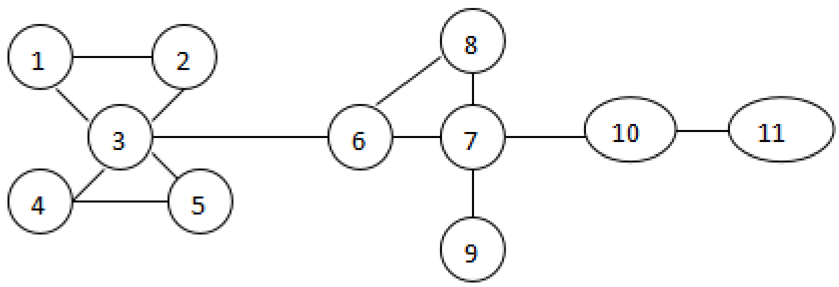
17) Explain dynamic programming with a multistage graph.

UNIT V

Explain BFS algorithm with example.

Explain DFS algorithm with example.

Use DFS for the given undirected graph starting at node 1



Explain Backtracking strategy for 8 Queens Problem.

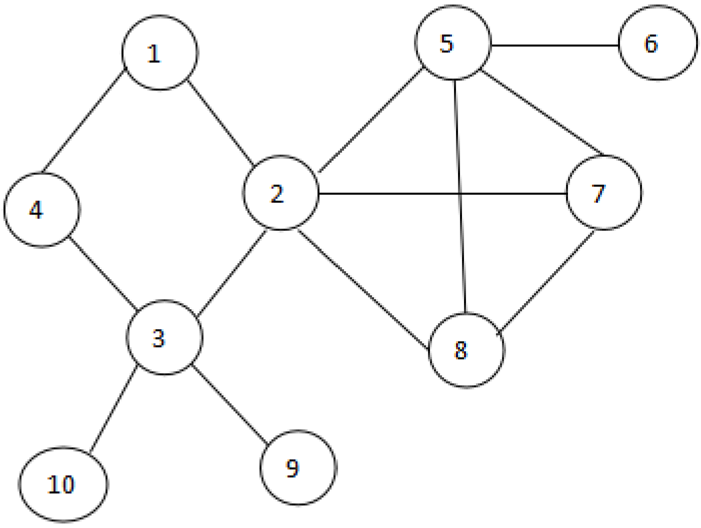
Explain M Colouring problem with an example.

Explain traversal of tree and write an algorithm to generate a tree given its in-order and post-order sequence.

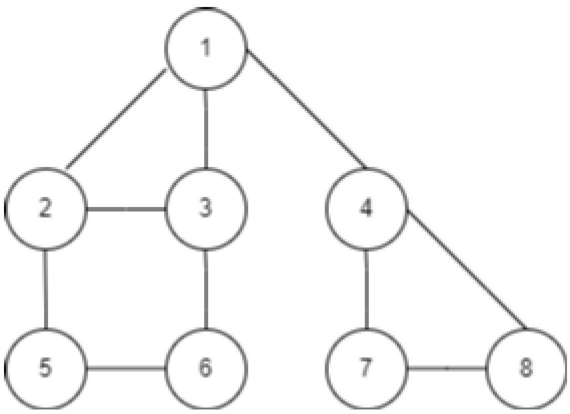
Explain backtracking framework in detail.

List and explain different state spaces.

Find the sequence of nodes traversed using DFS for the given undirected graph starting at node 1

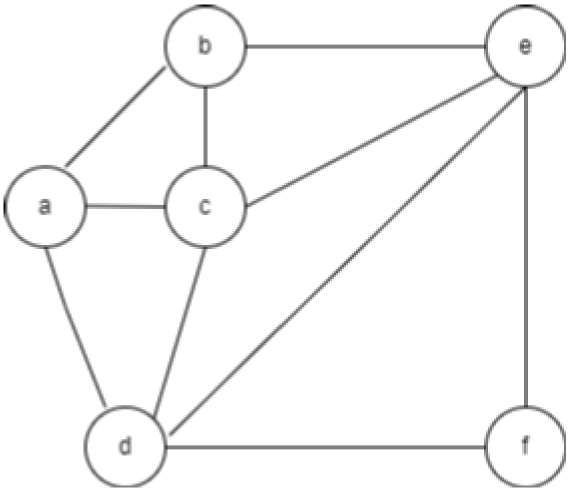


Explain DFS(Depth first search) algorithm. Simulate algorithm for following graph.



Explain Backtracking strategy for 4 Queens Problem.

Consider a graph G = <V,E> shown in following graph. Determine hamilton Circuit using backtracking strategy



Explain: a) N Queen Problem b)M-Colouring Problem c)Dead End

UNITVI

1)Explain polynomial time (P) algorithms.
2)Explain complexity of set operations and mapping.
3)Give the time analysis of algorithms ,explain using any search algorithm.
4)Explain efficiency of recursion with an example.
5)Explain complexity using the step counting principle.
6)Explain time complexity of algorithms.
7)Explain complexity and characteristics of the sorting algorithm.
8)Explain time complexity calculation for any sorting algorithm.
9)Explain time space trade off with the help of space efficient and time efficient solutions.
10) Explain worst case and average case behavior.
11) Explain the notion of complexity in brief.
12) Give a summary of complexity and characteristics of various sorting algorithms.
13) Explain in brief amortized analysis.
14)Explain time space trade off in algorithm research.
15)Explain Non Polynomial time (NP) algorithms.
16)Give the time analysis of algorithms as an example of matrix multiplication.